



Specialised Topics of Theoretical Physics

Majorana Modes in the Machine – Simulating Topological Phases with Quantum Circuits and AI

Error Taxonomy & Scaling to Large L

Edgar Harutyunyan
[3775171]

Supervisor: Prof. Dr. Bernd Rosenow

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LLM disclosure

This report was prepared with the assistance of OpenAI's GPT-5.6 Sol model, used with high reasoning effort through Codex. The model supported clarification and verification of nomenclature against the project implementation, navigation of the project code, consistency checking, and language editing of the manuscript. Its outputs were critically reviewed against the project code, numerical benchmarks, generated figures, and cited sources. All numerical data and figures reported here were generated by the project code. The author selected the final content and remains responsible for its accuracy.

1 Introduction

The *Majorana Modes in the Machine* project follows the Kitaev chain from its theoretical description to its simulation on noisy quantum circuits. G. Schewtschik's report, *Kitaev Chain & Bulk Topology*, introduces the bulk Hamiltonian, Bogoliubov–de Gennes spectrum, and topological phase diagram. Z. Shi's *Finite-Size Physics & Majorana Edge Modes* studies open chains, edge localisation, and Majorana hybridisation. J. Singh's *Qubit Encoding (Jordan–Wigner)* maps the fermionic model to qubits. Z. Liu's *Measuring Topology on Circuits (VQE + String Order)* develops the variational preparation and non-local edge-string measurement. D. Stoev's *NISQ Noise on the Diagnostic* investigates how realistic device noise degrades that measured signal.

In this report we focus on *Error Taxonomy & Scaling to Large L* . Here and throughout, L denotes the number of sites in the chain. First, we classify physical perturbations by mapping them to qubit operators and determine whether they preserve fermion parity and leave the bulk gap open. Next, we examine how the edge-string diagnostic scales using exact free-fermion references, fixed-depth variational states, and matrix-product-state (MPS) quantum trajectories under noise models derived from recorded IBM device calibrations up to $L = 100$. Thus, the two large-system limitations are insufficient circuit expressibility below a minimum depth r^* and noise accumulated over the entangling gates, whose number grows linearly with L . We identify the shallowest adequately expressive depth using exact small-system benchmarks and then test how the edge-string signal survives in calibrated large- L models. We discuss the protection principle and the large- L limit, while the channel-by-channel noise mechanics and noisy-optimisation results are treated in D. Stoev's *NISQ Noise on the Diagnostic*.

2 Theory

2.1 Protection criterion and diagnostics

Consider an open chain of L sites labelled $j = 0, \dots, L-1$. By open boundary conditions (OBC), we mean that there is no bond between sites $L-1$ and 0 . Under the Jordan–Wigner transformation, the chain is represented by the qubit Hamiltonian

$$H = -\frac{\mu}{2} \sum_{j=0}^{L-1} Z_j + \frac{t-\Delta}{2} \sum_{j=0}^{L-2} X_j X_{j+1} + \frac{t+\Delta}{2} \sum_{j=0}^{L-2} Y_j Y_{j+1}, \quad \hat{P} = \prod_{j=0}^{L-1} Z_j. \quad (2.1)$$

Here,

- μ is the chemical potential;
- t is the nearest-neighbour hopping amplitude;
- Δ is the real p -wave pairing amplitude;
- X_j , Y_j , and Z_j are Pauli operators acting on qubit j .

To connect Eq. (2.1) back to the fermionic variables, we use $c_j^\dagger$ and c_j for the creation and annihilation operators, respectively, and define $n_j = c_j^\dagger c_j$ as the occupation-number operator at site j . The code uses the convention $c_j^\dagger = (\prod_{k<j} Z_k) \sigma_j^+$ with $2\sigma_j^+ = X_j + iY_j$, where I is the single-qubit identity.

Hence $2n_j = I + Z_j$. Defining the total fermion number as $N = \sum_j n_j$, the fermion-parity operator is $P_f = (-1)^N = (-1)^L \prod_j Z_j$. Thus, the parity convention can be written as

$$P_f = (-1)^L \hat{P} = \begin{cases} \hat{P}, & L \text{ even}, \\ -\hat{P}, & L \text{ odd}. \end{cases}$$

For nonzero pairing, the bulk is topological for $|\mu| < 2|t|$. The bulk excitation gap Δ_{bulk} , defined as the minimum positive quasiparticle energy, closes at $|\mu| = 2|t|$ [3].

Let V denote a perturbation added to the clean Hamiltonian, so $H' = H + V$. We consider two perturbation families: $V_{\text{dis}} = \sum_j \delta\mu_j n_j$ with $\delta\mu_j \in [-W, W]$, and $V_{\text{poison}} = gX_0$. Thus, W and g set the respective disorder and poisoning strengths. Appendix A derives how these perturbations enter the Hamiltonian and change its eigenstates and measured observables. The perturbation V is *local* when it acts on one site or a finite set of neighbouring sites. It is parity even when $[V, \hat{P}] = 0$ and parity odd when $\{V, \hat{P}\} = 0$. We therefore require V to be *local*, *parity even*, and *weaker than* Δ_{bulk} . Such a perturbation may deform the wavefunction but cannot mix the logical sectors ($P_f = \pm 1$). Once its strength approaches the bulk gap, it can create locally trivial regions or close the bulk gap, thereby destroying the phase. A parity-odd perturbation instead changes the logical sector directly. The physical addition or removal of one fermion is called *quasiparticle poisoning*; increasing L does not suppress this logical error [6].

To define the measured diagnostic, we introduce the Majorana operators $\gamma_{2j} = c_j + c_j^\dagger = (\prod_{k<j} Z_k) X_j$ and $\gamma_{2j+1} = i(c_j - c_j^\dagger) = (\prod_{k<j} Z_k) Y_j$. The logical quantum number is the fermion parity P_f ($\hat{P}$ for even L), while the finite-chain edge diagnostic used in the simulations is

$$O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1} = -i\gamma_1 \gamma_{2L-2}. \quad (2.2)$$

This edge-string operator is a bilinear of the two boundary Majorana operators. Its expectation value is large for the prepared finite-chain state in the topological regime, whereas it is suppressed in the trivial regime. This makes it a useful finite-size diagnostic rather than an independent bulk topological invariant. A poisoning event always changes fermion parity, whereas its immediate effect on this particular string depends on the event's site and Pauli component. In summary, parity is the universal indicator of poisoning, whereas O_{edge} measures whether the prepared edge correlation survives.

2.2 Physical errors and emulator errors

Table 1 combines the physical and NISQ error dictionaries. Recall that protection requires an error to be local, parity even, and weaker than Δ_{bulk} .

Physical error	Jordan–Wigner qubit form	NISQ equivalent	Protection status
Charge noise $\delta\mu_i n_i$	Z_i field; parity even	T_2 ; coherent Z	ROBUST if sub-gap
Quasiparticle poisoning $c_i^{(\dagger)}$	$\frac{1}{2}(\prod_{k<i} Z_k)(X_i \mp iY_i)$; parity odd	T_1 damping	FATAL : sectors mix
Gap closure $ \mu \rightarrow 2 t $	Z field overtakes the XX/YY bonds	large coherent Z drift	FATAL : phase lost
Mixed Pauli: Z	Z_i ; parity even	depolarising Z branch	ROBUST if sub-gap
Mixed Pauli: X/Y	X_i, Y_i ; parity odd	depolarising X/Y branches	FATAL : sectors mix

Table 1: Taxonomy of physical perturbations and device noise.

The taxonomy transfers under Jordan–Wigner, but the transmon does not inherit the wire's gap protection. This emulator limitation is quantified by comparing the noisy and ideal versions of the same circuit:

$\epsilon_{\text{edge}} = |\langle O_{\text{edge}} \rangle_{\text{noisy}} - \langle O_{\text{edge}} \rangle_{\text{ideal}}|$. In Qiskit, two-qubit gate noise mixes parity-even charge/bond errors and parity-odd poisoning errors [5].

2.3 Scaling to large L

Large- L behaviour is governed by competing intrinsic and circuit scales. On the $t = \Delta$ line, the clean Majorana splitting in the topological regime obeys [3, 4]

$$\delta E(L) \simeq 2|t|(1 - \lambda^2)\lambda^L, \quad \lambda = \frac{|\mu|}{2|t|} = e^{-1/\xi}, \quad (2.3)$$

so increasing L exponentially suppresses edge-mode hybridisation. For a depth- r linear entangler, however, $n_{\text{CX}} = r(L - 1)$. A simple envelope for independent two-qubit errors therefore gives

$$|\langle O_{\text{edge}} \rangle_{\text{noisy}}| \sim |\langle O_{\text{edge}} \rangle_{\text{ideal}}| (1 - p_{\text{CX}})^{r(L-1)} \simeq |\langle O_{\text{edge}} \rangle_{\text{ideal}}| e^{-p_{\text{CX}} r(L-1)}. \quad (2.4)$$

The gate-level origin of this envelope and the $L = 4$ depth optimum are analysed in D. Stoev's report; here, we use the envelope only to study the dependence on L after fixing the shallowest adequately expressive depth. At fixed r , a nearest-neighbour circuit has bond dimension $\chi \leq 2^r$ and MPS memory $O(L\chi^2)$ [7]; see Appendix B. Thus, increasing L improves the intrinsic protection of the logical wire but simultaneously makes its circuit diagnostic more vulnerable, provided that the fixed depth remains expressive.

3 Methods and Results

3.1 Charge noise versus quasiparticle poisoning

We first test parity-even charge noise, modelled as local chemical-potential disorder $\mu_j = \mu_0 + U[-W, W]$. Since this channel preserves parity and the quadratic form of the Hamiltonian, a real antisymmetric $2L \times 2L$ Majorana matrix gives the ground-state energy, covariance matrix, parity gap, and Eq. (2.2) in $O(L^3)$ time [4]. The calculation used $t = \Delta = 1$, $\mu_0 = t$, $L = 16, 32$, 21 disorder strengths, and 300 realisations per point. Here the clean bulk gap is $\Delta_{\text{bulk}} = 2t - |\mu_0| = t$. Figure 1 shows that at $L = 16$ the disorder-averaged edge string changes only from 0.750 at $W = 0$ to 0.741 ± 0.115 at $W = 0.75$, before eroding more strongly near and above the gap.

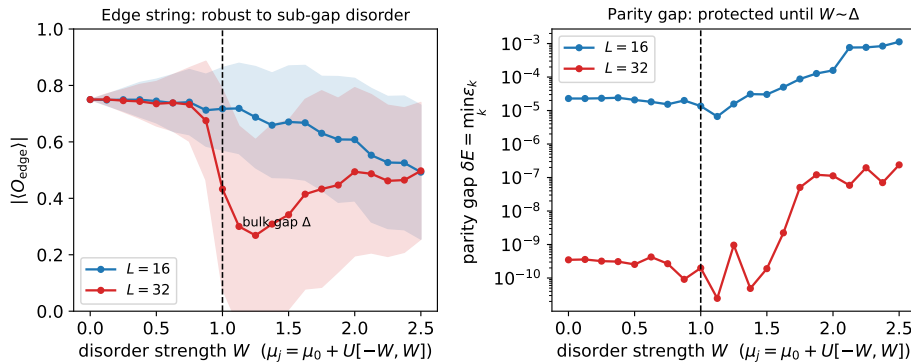


Figure 1: Disorder-averaged response to charge noise: edge string with one-standard-deviation bands (left) and parity gap (right).

Next, we test parity-odd quasiparticle poisoning using the static boundary term gX_0 , representing the left Majorana $c_0 + c_0^\dagger$. This Hamiltonian is no longer parity block diagonal, so a sparse exact diagonalisation of the full 2^{10} -dimensional qubit Hamiltonian was used. For a direct comparison, the Hamiltonian with the parity-even charge-noise Z field was solved in the even sector and averaged over 120 realisations. Figure 2 shows that charge noise keeps $\langle \hat{P} \rangle = 1$, whereas fermion parity departs continuously from one for any nonzero g . At a matched perturbation strength of 0.30, the edge strings under charge noise and poisoning are 0.750 and 0.614, respectively; at 0.78, the poisoning result has fallen to 0.318 with $\langle \hat{P} \rangle = 0.54$.

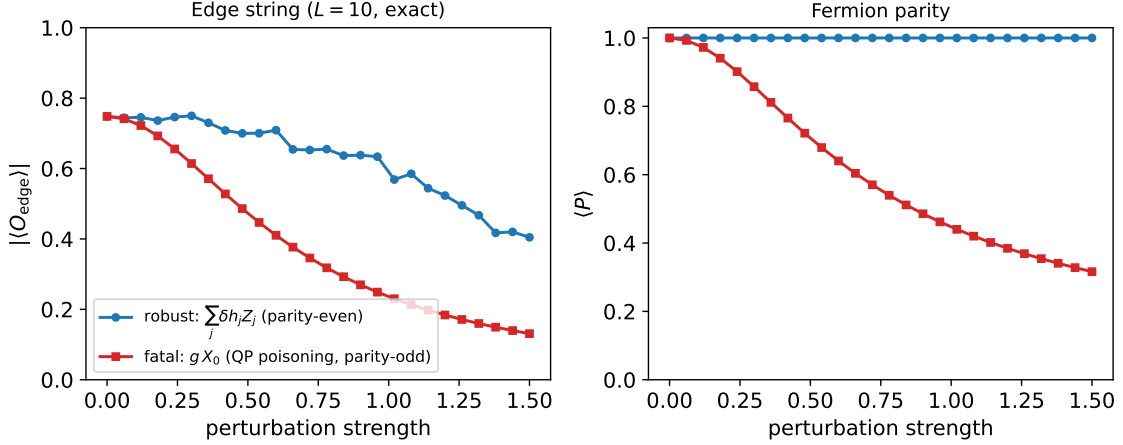


Figure 2: Responses to the two error channels: edge string (left) and fermion parity (right).

3.2 Large- L simulation

Before choosing a method, we separate three bottlenecks. Dense exact diagonalisation (ED) stores a $2^L \times 2^L = 4^L$ Hamiltonian, an ideal circuit simulation normally stores 2^L amplitudes in a statevector, and exact noisy simulation stores a 4^L density matrix. We address these bottlenecks as follows:

1. *Dense ED* $\rightarrow$ *free fermions*. A free-fermion Hamiltonian is quadratic in fermionic operators and diagonalises into independent Bogoliubov quasiparticles. A $2L \times 2L$ covariance calculation then gives the exact reference in $O(L^3)$ time. For $L = 4, 6, 8$, it agrees with dense qubit diagonalisation to about 10^{-15} .
2. *Statevector* $\rightarrow$ *MPS*. Through $L = 16$, sparse Pauli operators are evaluated on a statevector. At larger L , the same circuit state is stored as an MPS.
3. *Density matrix* $\rightarrow$ *MPS trajectories*. A Kraus trajectory is one pure-state realisation of the noisy circuit. Averaging many trajectories estimates the noisy expectation without storing the 4^L density matrix [7, 2].

At the gapped working point $\mu = 0.5t$, fixed-depth scans through $L = 16$ found a sharp expressibility threshold: $r = 2, 3$ failed beyond small chains, but $r^* = 4$ kept the noiseless edge string near the exact value $\langle O_{\text{edge}} \rangle \simeq 0.94$. Thus the depth needed for this observable did not grow over the tested range. With a deliberately harsh depolarising rate $p_{\text{CX}} = 0.05$, the edge signal fell from 0.681 at $L = 4$ to 0.074 ± 0.006 at $L = 16$; the recorded Cairo calibration ($p_{2q} = 7.53 \times 10^{-3}$) instead retained 0.523 ± 0.005 at $L = 16$.

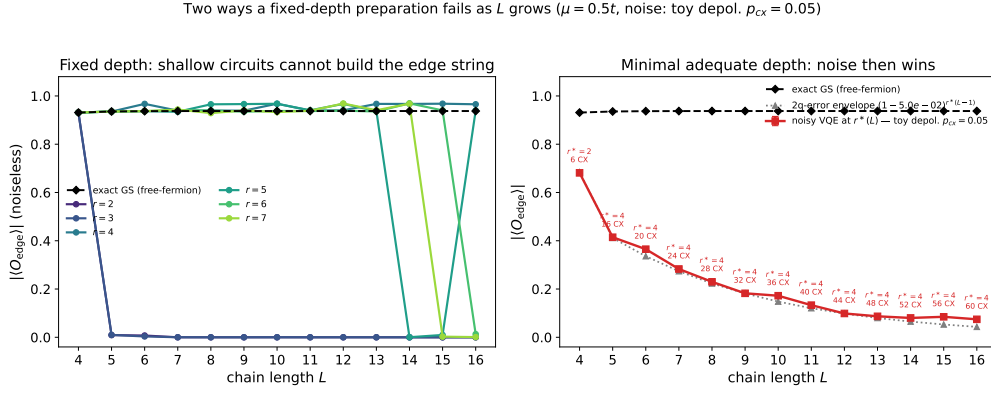


Figure 3: Noiseless edge string versus chain length and ansatz depth (left), and noisy edge string at the shallowest adequate depth (right).

The final sweep grew the chain in two-site steps, re-optimising the $r = 4$ parameters at each step, and used MPS at large L . These were simulations rather than hardware runs: recorded one-qubit noise and a median two-qubit noise channel were applied to the linear nearest-neighbour circuit without routing it onto the device's heavy-hex connectivity [1]. Each noisy point was obtained from 16,000 trajectories; the error bars show standard errors calculated from their variance. Computations for this work were done (in part) using resources of the Leipzig University Computing Center.

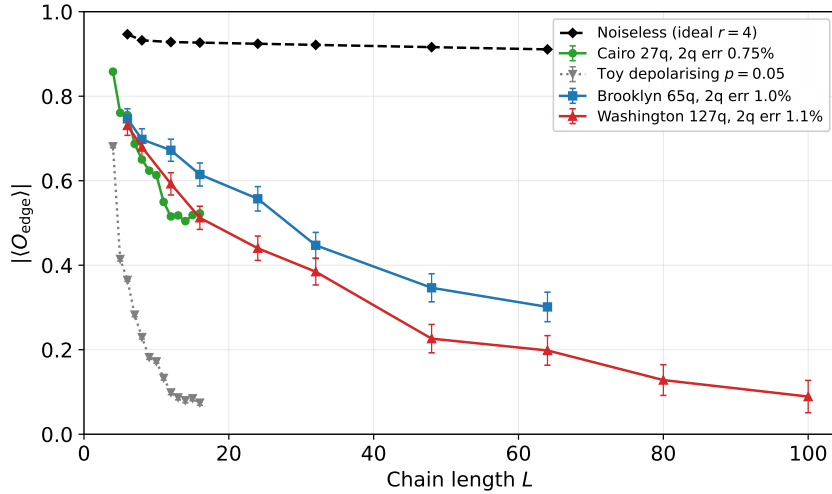


Figure 4: Fixed-depth edge-string expectations for ideal, toy-depolarising, and device-calibrated models; bars show trajectory standard errors.

The ideal signal is 0.947 at $L = 6$ and 0.899 at $L = 100$. At $L = 64$, Brooklyn (error 1.02×10^{-2}) gives 0.301 ± 0.035 , whereas Washington (error 1.15×10^{-2}) gives 0.198 ± 0.035 and reaches 0.089 ± 0.038 at $L = 100$.

Recall that a projective measurement of the Pauli string yields $+1$ or -1 , with $\langle O_{\text{edge}} \rangle = P(+1) - P(-1)$. Thus, the value 0.089 at $L = 100$ corresponds to 54.45% positive and 45.55% negative outcomes, whereas the ideal contrast is 0.899. Although the Hamiltonian remains in the topological regime, decoherence accumulated over the growing gate count leaves the edge signal barely readable and constitutes the dominant large- L simulation wall.

4 On the LLM collaboration

To keep this report compact, we avoid repeating material available on the LLM-feedback project page. Instead, Figure 5 illustrates a representative correction made during the collaboration. The model initially inferred that the required ansatz depth increased with L . After receiving the extended cluster results for $L = 4, \dots, 16$, it revised this claim: $r^* = 4$ remained sufficient over the tested range, while the large- L decay arose primarily from the growing CNOT count. It also identified the anomalous high-depth points as optimisation failures rather than physical effects.

This is the **toy-model run** ($p_{\text{cx}}=0.05$) extended to $L=4..16$, reps family 2..7, on the cluster (job 25357569). Here's what it says.

The headline: the two failure modes *decouple* – and it's noise, not expressibility

Panel A – expressibility is cheap and L-independent. A constant depth $r^*=4$ reproduces the edge string all the way to $L=16$ (the teal $r=4$ line sits flat at ~ 0.94 , on top of the free-fermion truth). $r^*(L) = [2, 4, 4, 4, 4, 4, 4]$ – it does **not** grow with L . That's physical: the edge Majoranas are *localized* at the two ends, so building them costs $\sim$ constant depth regardless of how long the chain between them is. Shallow $r=2,3$ still fail beyond $L=4$ (they can't form the correlation at all), but once you clear the depth threshold, it holds at any L .

This actually **corrects** the framing in my docstring/notes ("the threshold grows with L ") – the better-optimized large- L data says the threshold is **flat at 4**.

Panel B – noise is the entire large- L story. Because r^* is fixed at 4, the CNOT count grows *linearly*, $n_{\text{cnot}} = 4(L-1)$: $6 \rightarrow 20 \rightarrow 28 \rightarrow 36 \rightarrow 44 \rightarrow 52 \rightarrow 60$. So the toy-noise edge decays exponentially in L : $0.68 \rightarrow 0.37 \rightarrow 0.23 \rightarrow 0.17 \rightarrow 0.10 \rightarrow 0.08 \rightarrow 0.07$, tracking the $(1-p)^{n_{\text{cnot}}}$ envelope. By $L=16$ it's ~ 0.07 – within a couple of error bars of the noise floor. Under this noise, the diagnostic is effectively dead by $L=14-16$.

Two important caveats

1. **This is the toy $p=0.05$ – $\sim 6\times$ harsher than real hardware.** The same 60 CNOTs under real Cairo (2q err 0.0075) give $(1-0.0075)^{60} \approx 0.64$, i.e. edge ≈ 0.6 at $L=16$ – it would survive. So "noise kills it by $L=16$ " is an artifact of the toy strength. The device-tagged run (`BACKEND=FakeCairoV2`) is the one that shows the honest reach; worth running to $L=16$ for the real conclusion.
2. **Three optimizer glitches in Panel A** – the V-shaped dips to ~ 0 : $r=5$ at $L=14$, $r=6$ at $L=16$, $r=7$ at $L=16$. These are *not* physics – a lower depth ($r=4$) succeeds at those same L – they're under-optimization at high parameter counts ($L=16, r=7 = 128$ params, only 8 starts). They don't touch Panel B (which always uses the converged $r^*=4$), but they're visible blemishes.

Figure 5: The Opus 4.8 model revising its initial scaling interpretation after comparison with the extended cluster data.

A How a perturbation changes the Hamiltonian and states

This appendix makes explicit how the perturbation operator V used in Sec. 2 enters the Hamiltonian and changes its eigenstates, Majorana wavefunctions, and measured observables.

From a physical parameter to an operator. Write the clean Hamiltonian as H_0 and the perturbed Hamiltonian as

$$H' = H_0 + V, \quad V = \sum_{\alpha} \lambda_{\alpha} O_{\alpha}, \quad (\text{A.1})$$

where O_{α} specifies how the environment couples to the chain and λ_{α} is its strength. For chemical-potential disorder $\mu_j = \mu_0 + \delta\mu_j$, the change relative to Eq. (2.1) is

$$V_{\text{dis}} = -\frac{1}{2} \sum_j \delta\mu_j Z_j = -\sum_j \delta\mu_j \left(n_j - \frac{I}{2} \right), \quad \delta\mu_j \in [-W, W]. \quad (\text{A.2})$$

The identity contribution has no effect on eigenvectors. If the disorder coefficient is instead defined as $h_j = -\delta\mu_j$, this becomes $\sum_j h_j n_j$ up to that constant, which is the shorthand used in the taxonomy. The boundary poisoning term is instead

$$V_{\text{poison}} = gX_0 = g(c_0 + c_0^{\dagger}). \quad (\text{A.3})$$

Consequently $[V_{\text{dis}}, \hat{P}] = 0$, whereas $\{V_{\text{poison}}, \hat{P}\} = 0$.

Change of a many-body wavefunction. Let $H_0|n\rangle = E_n|n\rangle$. For an isolated, non-degenerate eigenvalue, first-order stationary perturbation theory gives

$$E'_n = E_n + \langle n|V|n\rangle + O(V^2), \quad (\text{A.4})$$

$$|n'\rangle = |n\rangle + \sum_{m \neq n} |m\rangle \frac{\langle m|V|n\rangle}{E_n - E_m} + O(V^2). \quad (\text{A.5})$$

Thus V changes a wavefunction only through matrix elements $\langle m|V|n\rangle$, while the energy denominators suppress mixing with states separated by the bulk gap.

Selection by fermion parity. For parity eigenstates $\hat{P}|n, p\rangle = p|n, p\rangle$, with $p = \pm 1$, define $M_{qp} = \langle m, q|V|n, p\rangle$. The commutator or anticommutator with $\hat{P}$ implies

$$\begin{aligned} [V, \hat{P}] = 0 : & \quad (q - p)M_{qp} = 0, \quad \text{only equal parities can mix,} \\ \{V, \hat{P}\} = 0 : & \quad (q + p)M_{qp} = 0, \quad \text{only opposite parities can mix.} \end{aligned} \quad (\text{A.6})$$

For the nearly degenerate topological states $|e\rangle$ and $|o\rangle$, the appropriate degenerate-subspace Hamiltonian is

$$H_{\text{eff}} = \begin{pmatrix} E_e + v_{ee} & v_{eo} \\ v_{oe} & E_o + v_{oo} \end{pmatrix}, \quad v_{pq} = \langle p|V|q\rangle. \quad (\text{A.7})$$

A parity-even perturbation has $v_{eo} = v_{oe} = 0$ and therefore cannot mix the two sectors. Symmetry alone does not require $v_{ee} = v_{oo}$; locality and the bulk gap suppress their difference when the two edge modes

are well separated. For a parity-odd perturbation, $v_{ee} = v_{oo} = 0$ but the off-diagonal element is allowed. Its two energies are

$$E_{\pm} = \frac{E_e + E_o}{2} \pm \sqrt{\left(\frac{E_e - E_o}{2}\right)^2 + |v_{eo}|^2}. \quad (\text{A.8})$$

Even when the clean finite-size splitting $|E_e - E_o|$ is exponentially small, a permitted poisoning matrix element can therefore mix and split the sectors.

Majorana profiles and observables. For a quadratic, parity-even perturbation, the Bogoliubov–de Gennes (BdG) matrix changes as $h'_{\text{BdG}} = h_{\text{BdG}} + \delta h$. If $h_{\text{BdG}}|u_n\rangle = \varepsilon_n|u_n\rangle$ and the level is isolated, its spatial wavefunction changes by

$$|\delta u_n\rangle = \sum_{m \neq n} |u_m\rangle \frac{\langle u_m | \delta h | u_n \rangle}{\varepsilon_n - \varepsilon_m}. \quad (\text{A.9})$$

This is how sub-gap disorder deforms the left- and right-edge Majorana profiles without necessarily destroying them. The nearly degenerate edge pair itself must be treated with Eq. (A.7) rather than the non-degenerate formula.

Finally, for an observable O in a non-degenerate state $|0\rangle$, the first-order response is

$$\delta\langle O \rangle = 2 \operatorname{Re} \sum_{m \neq 0} \frac{\langle 0 | O | m \rangle \langle m | V | 0 \rangle}{E_0 - E_m} + O(V^2). \quad (\text{A.10})$$

Taking $O = O_{\text{edge}}$ explains why parity-even disorder can continuously deform the measured edge string even while parity remains exactly fixed. A parity-odd term instead mixes the two parity sectors through Eq. (A.7), so neither $\langle \hat{P} \rangle$ nor the edge-string signal is protected.

B Fixed-depth MPS memory scaling

Consider a cut between sites b and $b + 1$. Its Schmidt decomposition is

$$|\psi\rangle = \sum_{\alpha=1}^{\chi_b} s_{\alpha} |\alpha\rangle_{0\dots b} |\alpha\rangle_{b+1\dots L-1}, \quad (\text{B.1})$$

Here χ_b is the bond dimension. A product input has $\chi_b = 1$, and single-qubit rotations leave it unchanged. A crossing CNOT has operator Schmidt rank two, $\text{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$, so it at most doubles χ_b . Since each linear layer crosses the cut once, r repetitions give $\chi_b \leq 2^r$ and hence $\chi = \max_b \chi_b \leq 2^r$.

The tensor at site j has dimensions $\chi_{j-1} \times 2 \times \chi_j$. The total number of stored complex entries is therefore bounded by

$$N_{\text{MPS}} = \sum_{j=0}^{L-1} 2\chi_{j-1}\chi_j \leq 2L\chi^2 = O(L\chi^2). \quad (\text{B.2})$$

For $r = 4$, $\chi \leq 16$, so the memory is linear in L . Kraus trajectories store pure-state MPS branches rather than the 4^L density matrix; general two-site noise changes the fixed-depth bond constant to $2^{O(r)}$, not the linear dependence on L .

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